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1.0 Introduction

Academic planning involves choosing in between a set of objectives and a set of constraints. Students are also supposed to select modules that are not only satisfactory to the programme requirements, but also to their interests and time limitations. This complicates the decision problem to a complex problem that cannot be effectively tackled using intuition particularly where two or more constraints are intertwined with one another.

This report is aimed at a case of Sophie who is a final year student of Finance and has to take five modules in her academic year. The decision has several institutional requirements. Specifically, Sophie will need to select two mandatory courses, i.e. Business Strategy (BIS101) and International Finance (FIN300), one industry-specific course, and two courses elective in finance. In addition to these criteria, Sophie is interested in attaining optimum satisfaction based on her interest in the contents in modules, the quality of lecturers and convenience of the schedule.

The problem is also very complicated due to scheduling restrictions. There are modules that cannot be taken at the same time as they run on time even on an overlapping time slot. Therefore, Sophie ought to ensure that the selected modules do not bring a clash in time schedules. Also, there is only one time to take modules and a module can only be assigned to a particular time slot.

Using these restrictions, the problem is an optimisation problem formulated. These issues should be solved with the help of the most suitable approach Binary Integer Programming (BIP) which helps to code the decisions to binary form (i.e. 1/0). This will ensure that every constraint is closely monitored and the combination of modules will be the best in bringing out the highest fulfillment of Sophie.

This report is addressed to solving this decision problem by Binary Integer Programming model and regarding the applicability of obtained solution.

2.0 Literature Review

Operations Research (OR) is a discipline, which applies mathematical modelling, analytical mechanisms and optimisation structures in decision-making. It is particularly relevant when decisions are required to be made under a limited setting and decisions about conflicting objectives are required.

Integer Programming (IP) is one of the branches of OR and is used widely as far as discrete decision variables are concerned. According to Panneerselvam (2023), IP would be particularly fitting, in cases where the decision-process is binary, and allocation and selection are used. The decision variables in these problems are integer valued only, and can be constrained to 0 or 1 and so the method is thus appropriate in depicting yes-or-no choices.

Binary Integer Programming as Carter, Price and Rabadi (2018) emphasize is quite handy in the actual life scenario such as scheduling, resource allocation, and assignment. These types of problems are to select part of the choices that is offered to one and satisfy a number of constraints, which is quite close to the module selection problem of Sophie.

The new studies have also determined the greater importance of OR in dealing with the intricate decision making issues. Govindan (2025) notes the possibility of introducing advanced optimisation techniques in order to maximise efficiency and quality of decisions in a multi constraint system. On the same note, Acquaye (2026) introduces the problem of the application of OR to address sustainability and operation issues whereby an optimisation model could be efficient in managing conflicting objectives and making them feasible.

An example of an application of the method of Integer Programming is in timetabling problems, which are to distribute activities to time slots in a conflict-free fashion. Such problems normally entail constraints such as resource constraints, time constraints, and selection constraints. The absence of rigidity in IP models provides an opportunity to combine these constraints into a single model and find the most optimal solutions.

The problem of Sophie is especially appropriate to be solved with the help of the Binary Integer Programming because it allows:

- Modelling the decision of the module selection as a binary variable.

- Combination of academic and time restriction.
- Application of a preference-based objective to be optimised.

Therefore, the BIP application provides a methodological and objective way of fixing the problem of module selection.

2.1 Justification for the Use of Binary Integer Programming

Besides being applicable in general, Binary Integer Programming is especially appropriate to the problem of Sophie because of the discrete and mutually exclusive character of the decisions she is forced to make. Each module selection is a binary decision, either a module is selected or not, and continuous optimisation methods are therefore unsuitable.

Also, the existence of numerous constraints such as mandatory requirements, selection constraints, and time constraints demand a framework of models that can tackle logical requirements at the same time. Binary Integer Programming enables such conditions to be defined mathematically, which is not ambiguous and without these constraints, all constraints are followed strictly during optimisation process.

The other major benefit of Binary integer Programming is that it ensures global optimality given that the conditions are well defined. As opposed to heuristic or trial-and-error methods where methods can result in sub-optimal or inconsistent decisions, the optimisation model analyses every possible combination of them to derive the best possible solution.

This may be especially useful in the context of academic planning since it eliminates subjectivity and offers a logical foundation of decision-making. As such, the Binary Integer Programming is not only suitable but also necessary in the determination of a reliable and optimal solution in the case of Sophie in the selection of modules.

3.0 Input Data Handling

3.1 Sets

The model is defined using two primary sets:

$i = 1,2,\ldots,10$ representing the available time slots

j = 1,2,…,8 representing the available modules

The modules considered in this study are:

- BIS101 (Business Strategy)
- FIN300 (International Finance)
- CIS101 (Business Computing)
- CIS102 (Web Design for Nonprofit Organisations)
- FIN315 (Data Analysis in Finance)
- FIN316 (Risk Management)
- FIN317 (Options, Futures and Swaps)
- FIN318 (Fixed Instruments and Markets)

3.2 Parameters

The key parameter used in the model is:

$$R_{ij}$$

which represents the rating assigned to module j in time slot i.

These ratings are based on Sophie's preferences and reflect a combination of:

- Interest in module content
- Lecturer effectiveness
- Suitability of the time slot

The rating scale will be 0 to 5 where high value will be the preference. The rating is set to zero in case the module is not among the possible ones in a specific time slot. This is to ensure that the model will avoid a combination of module-time that is not available. The objective function is founded on the rating matrix that is highly valuable in the process of achieving the optimum solution.

3.3 Data Structure and Preparation

The information fed was in a table format and was typed in Microsoft Excel. Time slots are placed in a row and modules in a column. Each cell of the table will contain the rating R_{ij} that will be utilized in the process of optimisation.

4.0 Model Formulation

The problem is formulated as a Binary Integer Programming model, where the objective is to maximise total satisfaction while satisfying all constraints.

4.1 Decision Variables

$$x_{ij} = \begin{cases} 1 \; if \; module \; j \; is \; selected \; in \; a \; time \; slot \; i, \\ 0 \; otherwise \end{cases}$$

Where:

$$x_{ij} \in \{0,1\}$$

4.2 Objective Function

Maximise total satisfaction:

$$Maximixe \; z = \sum_{i=1}^{10} \sum_{j=1}^{8} R_{ij} X_{ij}$$

4.3 Constraints

(1) Total modules

$$\sum_{i=1}^{10} \sum_{j=1}^{8} Xij = 5$$

(2) BS101 and FIN300 are core modules and must be taken.

$$\sum_{i=1}^{10} X_{ij} = 1 \qquad for \; j = 1,2$$

(3) Industry module

$$\sum_{i=1}^{10} X_{i3} + \sum_{i=1}^{10} X_{i4} = 1$$

(4) Finance optional modules

$$\sum_{i=1}^{10}(x_{i5} + x_{i6} + x_{i7} + x_{i8}) = 2$$

(5) No time clash and uniqueness

$$\sum_{i=1}^{10}\sum_{j=1}^{8} Xij \leq 1$$

(6) Monday/Wednesday Clash

Slot 6 and 7 overlap

$$\sum_{j=1}^{8} x_{6j} + \sum_{j=1}^{8} x_{7j} \leq 1$$

Slot 7 and 8 overlap

$$\sum_{j=1}^{8} x_{7j} + \sum_{j=1}^{8} x_{8j} \leq 1$$

(7) Binary restriction

$$x_{ij} \in \{0,1\}$$

4.4 Assumptions of the Model

To reduce the number of variables in the model so as to make them manageable, a few assumptions were made during the formulation of the model.

First, it is presumed that the ratings linked to each of the modules give a proper representation of what Sophie would prefer. These rating are assumed to be fixed and reliable yet in the actual sense, preferences can change in the long run or can be dependent on other factors which are not captured in the model.

Second, the model fails to consider the relative difficulty or the workload in each module. It presupposes that all modules are equally manageable, which is not possible to be real in the academic environment.

Lastly, the model is based on the assumption that, all the factors that were relevant in deciding on behalf of Sophie have been represented in the rating system. The external influences (peer influence, career goals, or future specialisation) are not specifically mentioned.

The assumptions are required to have a manageable model structure but there are limitations that should be taken care of in the interpretation of the results.

5.0 Model Implementation

The solution of the Binary integer Programming model was obtained through Microsoft Excel Solver which is an easy to use optimisation tool that are normally used to solve the linear and integer programming problems. It is highly flexible and simple and thus is specifically applicable in academic management such as in this module selection problem.

5.1 Model Setup in Excel

		1	2	3	4	5	6	7	8
	Time Slots	BIS101	FIN300	CIS 101	CIS 102	FIN 315	FIN 316	FIN 317	FIN 318
1	M Eve	4.3					3.6		3
2	T Eve	3.8			3.7			3.2	
3	W Eve	3.5	3.5						3.5
4	Th Eve					3			
5	F Eve	3.5							
6	M 1:25-3:15W 1:25-2:20						3.9		
7	M 1:25-2:2	4.6				3.7			
8	W 2:30-5:15			4.4	3.5				
9	T 1:25-3:1	2.7	3.3					3.4	
10	Th 2:30-5:15			3.1					

Figure 1: Excel Model Setup

This model had been developed in the form of a matrix where:

- Rows represent time slots (i = 1 to 10)
- Columns represent modules (j = 1 to 8)

- Each cell corresponds to a decision variable X_{ij}

A corresponding matrix of ratings R_{ij} was created to store Sophie's preference scores. The decision variable matrix was initialised with binary values (0 or 1), representing whether a module is selected in a given time slot.

5.2 Objective Function Implementation

The objective function was entered using the SUMPRODUCT function in excel:

Z = SUMPRODUCT(R_{ij}, X_{ij})

This function multiplies each rating by its corresponding decision variable and sums the results, producing the total satisfaction score.

5.3 Constraint Implementation in Solver

All the mathematical constraints were entered into excel in their form and entered in Solver as:

- Total module constraint: Sum of all decision variables equals 5
- Core modules constraint: Column sums for BIS101 and FIN300 equal 1
- Industry module constraint: Combined sum of CIS101 and CIS102 equals 1
- Finance modules constraint: Sum of finance module selections equals 2
- Scheduling feasibility:

i. Row sums ≤ 1 (no time clashes)

ii. Column sums ≤ 1 (no repeated modules)

- Overlapping constraints: There are more constraints between time slot 6 and 7, and 7 and 8.

5.4 Solver Configuration

Solver was configured to the settings as follows:

- Objective: Maximise total satisfaction (Z)
- Variable cells: Decision variable matrix
- Constraints: All linear constraints stipulated above
- Variable type: Binary

The Simplex LP solving with integer restrictions was used to solve it. Solver applies branch-and-bound techniques in solving binary problems although it is a linear problem solver.

5.5 Model Validation

After solving, the results were carefully validated to ensure:

- All constraints were satisfied
- No module was selected more than once
- No overlapping time slots were chosen
- The total number of modules selected was exactly five

This validation step was essential to confirm that the solution was both optimal and feasible.

6.0 Results and Evaluation

6.1 Optimal Solution

The final solution obtained from Solver produced an optimal satisfaction score of:

Z = 19.5

This represents the maximum achievable satisfaction under all given constraints, including timetable feasibility.

6.2 Interpretation of the Solution

Slot	BIS101	FIN300	CIS 101	CIS 102	FIN 315	FIN 316	FIN 317	FIN 318	Obj
1	1	0	0	0	0	0	0	0	19.5
2	0	0	0	0	0	0	0	0	
3	0	1	0	0	0	0	0	0	
4	0	0	0	0	0	0	0	0	
5	0	0	0	0	0	0	0	0	
6	0	0	0	0	0	1	0	0	
7	0	0	0	0	0	0	0	0	
8	0	0	1	0	0	0	0	0	
9	0	0	0	0	0	0	1	0	
10	0	0	0	0	0	0	0	0	

Figure 2: Final Result of the Model

The modules selected are in line with the requirements in terms of timeline and the academics:

- Exactly five modules were chosen
- Both of the mandatory courses (BIS101 and FIN300) were accomplished.
- One industry module was selected
- Two finance optional modules were selected
- Time slot with more than one module did not exist.
- No overlapping time slots were selected

This confirms that the model successfully identified a **feasible and optimal module combination**.

6.3 Comparison with Initial Model

Slot	BIS101	FIN300	CIS 101	CIS 102	FIN 315	FIN 316	FIN 317	FIN 318	Obj
1	0	0	0	0	0	1	0	0	19.8
2	0	0	0	0	0	0	0	0	
3	0	1	0	0	0	0	0	0	
4	0	0	0	0	0	0	0	0	
5	0	0	0	0	0	0	0	0	
6	0	0	0	0	0	1	0	0	
7	1	0	0	0	0	0	0	0	
8	0	0	1	0	0	0	0	0	
9	0	0	0	0	0	0	0	0	
10	0	0	0	0	0	0	0	0	

Figure 3: Initial Result before Adding all the Constraint

Before introducing the overlapping time constraints, the model produced a higher objective value:

Z = 19.8

However, this solution included timetable conflicts, making it impractical. After adding the Monday/Wednesday overlap constraints, the objective value reduced slightly to:

Z = 19.5

6.4 Trade-Off Analysis

This reduction leads to a considerable concept of optimisation; the trade-off between feasibility and optimality.

- The initial solution (Z = 19.8) was mathematically optimal but unrealistic

- The revised solution (Z = 19.5) is slightly lower but fully feasible

This shows that real-world constraints should also be included even though it may lead to some small objective value drop. The infeasible solution can never be of any use in comparison to a feasible solution.

6.5 Strengths of the Model

The model demonstrates several key strengths:

- Structured decision-making: Provides a clear and logical framework
- Optimality: Guarantees the best possible solution under constraints
- Flexibility: Can easily incorporate additional constraints
- Efficiency: Eliminates manual trial-and-error processes
- Scalability: Can be extended to larger problems

6.6 Limitations of the Model

Despite its strengths, the model has some limitations:

- Subjectivity of ratings: Preferences may not be entirely accurate
- Static assumptions: Does not account for changes over time
- Exclusion of workload factors: Difficulty level of modules is not considered
- Simplified constraints: Real-world scheduling may involve more complexities

These limitations suggest that while the model is effective, it could be further improved with additional data and considerations.

6.7 Practical Implications of the Results

The results obtained from the model have important practical implications for academic decision-making. By providing a structured and quantitative approach, the model enables students to make informed choices that align with both institutional requirements and personal preferences.

In real-world academic environments, students often rely on informal advice or subjective judgment when selecting modules. This can lead to suboptimal decisions, such as choosing modules that conflict in schedule or do not maximise personal satisfaction. The application of an

optimisation model addresses these issues by ensuring that all constraints are systematically considered.

Additionally, the model demonstrates how small changes in constraints can significantly impact the final outcome. The reduction in the objective value from 19.8 to 19.5 illustrates how incorporating realistic conditions can alter the optimal solution. This highlights the importance of accurately modelling real-world conditions when applying optimisation techniques.

7.0 Reflection

It was also made known during the modelling process, the strengths and the limitations of using optimisation techniques in the decision making process. The initial issue which was among the most severe failures was the absence of the module uniqueness constraint in the first place. The result of this was a non-solution whereby the same module was selected on repeat occasions. This error proved that one should be very careful with defining constraints to make them reflect the reality.

Another challenge was identifying hidden scheduling conflicts. Initially, the model did not account for overlapping time slots (6–7 and 7–8). The inclusion of these constraints improved the realism of the model and ensured that the final timetable was feasible.

This experience highlighted several important lessons:

7.1 Importance of Accurate Modelling

Even small errors in constraint formulation can lead to invalid solutions. Therefore, it is essential to:

- Clearly define all constraints
- Check both row and column restrictions
- Validate model outputs thoroughly

7.2 Role of Constraints in Optimisation

The constraints are not only limits, but also form the problem structure. The practical applicability of the solution is augmented by the introduction of realistic constraints, although at the cost of the objective value.

7.3 Learning from Iteration

The modeling process is recursive in nature. The first models can be considered unfinished, and enhancements are necessary. This is through a refining and refining process which results in improved and more precise solutions.

7.4 Application of Integer Programming

This coursework revealed that Binary Integer Programming is an effective decision solver to structured decision problems. It enables combination of several constraints and finds optimal solutions at an efficient level.

8.0 Conclusion

This report has shown how Binary Integer Programming is applicable when solving a module selection problem which is a real-world problem. The decision problem of Sophie could be addressed using an optimisation model, which then allowed to search through all possible combinations in a systematic way and determine the best solution. The solution that was arrived at scored a satisfaction of: Z= 19.5 and within all the academic and time schedule requirements. The presence of the time constraints that overlapped each other made the solution not only optimal but realistic and feasible. The paper emphasizes the relevance of mathematical modeling with practical deliberations. Although optimisation techniques may offer effective solutions, the degree to which they can be effective depends on how far the model can be realistic. On the whole, this coursework shows that Operations Research is a solid and trusted model to address complicated problems in decision making, especially, when it comes to the cases with multiple constraints and conflicting goals.

References

Acquaye, A. (2026) ‘Operational research for sustainability: a synthesis of methods, applications and challenges’, *Journal of the Operational Research Society*, 77(1), pp. 8–42.

Carter, M., Price, C.C. and Rabadi, G. (2018) *Operations Research: A Practical Introduction*. Boca Raton: CRC Press.

Govindan, K. (2025) ‘Applications of advances in operations research methods to achieve sustainable development goals (SDGs) in a supply chain management environment: review and future direction of research’, *Annals of Operations Research*, 349(2), pp. 425–450.

Panneerselvam, R. (2023) *Operations Research*. New Delhi: PHI Learning.